P A T E L M A R T B . V .

DemandIQ NL

Supply Chain Analytics Report

Predictive Inventory Intelligence & Demand Forecasting

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1. Executive Summary

Patel Mart B.V. is a Utrecht-based regional FMCG distributor supplying 10 retail locations across the Netherlands with €27.30 million in annual revenue. Prior to this project, inventory decisions were managed manually using a flat 14-day safety stock rule applied uniformly across all 3,000+ SKUs — regardless of demand variability, product category, or lead time. This approach resulted in €2.37 million in identifiable excess inventory and an estimated €593,000 in avoidable annual holding costs.

DemandIQ NL is a seven-phase end-to-end predictive analytics pipeline built to replace manual inventory intuition with data-driven intelligence. The project spans data engineering, demand forecasting, inventory risk scoring, and executive business intelligence — delivered via a live Power BI dashboard connected directly to a PostgreSQL data warehouse.

€2.37M	€593K	3,071	22.7%
Excess inventory identified	Annual holding cost saving	HIGH-risk SKU × Store flagged	Forecast error reduction (RMSE)

The forecasting engine benchmarked XGBoost against Prophet on 10 Class A SKUs from the highest-revenue store (CA_3 Utrecht). XGBoost outperformed Prophet on all 10 SKUs, achieving a 22.7% reduction in RMSE and 21.3% in MAPE. The inventory engine subsequently assessed 23,311 SKU × Store combinations, identifying 3,071 as HIGH risk — primarily concentrated in the Voeding (Food) category at a 15.64% risk rate.

This report documents the methodology, findings, challenges encountered, and recommendations for operational implementation and further development.

2. Introduction

2.1 Company Background

Patel Mart B.V. is a regional fast-moving consumer goods (FMCG) distributor headquartered in Utrecht, Netherlands. The company operates as an intermediary between product manufacturers and retail outlets, managing the procurement, storage, and delivery of goods across 10 retail locations spanning nine Dutch provinces: Utrecht, Zuid-Holland, Noord-Holland, Drenthe, Groningen, Zeeland, Gelderland, Overijssel, and Friesland.

With an annual revenue of €27.30 million and a portfolio exceeding 3,000 active SKUs across four product categories — Voeding (Food), Huishouden (Household), Hobby & Vrije Tijd (Hobbies & Leisure), and Persoonlijke Verzorging (Personal Care) — Patel Mart operates at a scale where manual inventory management is neither efficient nor reliable.

2.2 Project Context

DemandIQ NL was initiated by the data analytics function at Patel Mart in response to recurring operational challenges: stockouts on high-velocity food products, excess inventory in slow-moving categories, and the absence of any systematic mechanism for predicting demand or identifying risk before it materialises. The project was designed as a portfolio-grade end-to-end analytics system, built on real organisational data and developed to international supply chain analytics standards.

2.3 Report Objectives

This report serves three purposes:

- To document the analytical methodology and design decisions made across all seven project phases.
- To present the key findings from data exploration, forecasting, and inventory risk analysis.
- To provide evidence-based recommendations for operational implementation and future capability development at Patel Mart.

2.4 Data Sources

The underlying dataset is derived from the M5 Forecasting Competition (Kaggle), a widely used real-world retail demand benchmark employed in academic research and industry supply chain modelling. The original US Walmart retail dataset was systematically transformed to reflect the Dutch FMCG distribution context of Patel Mart B.V.:

- Product categories remapped to Dutch equivalents (Voeding, Huishouden, Hobby & Vrije Tijd, Persoonlijke Verzorging).
- Store locations reassigned to Dutch provinces and cities (Utrecht as anchor, plus nine additional provincial locations).
- Dates realigned to the Dutch retail calendar with public holiday flags applied: Koningsdag, Bevrijdingsdag, Sinterklaas, and Kerst.
- All pricing retained in euros (€) consistent with Dutch market conventions.

The dataset after transformation comprised 58.3 million rows across three staging tables: sales transactions (58.3M rows), sell prices, and the Dutch calendar.

3. Problem Statement

3.1 Industry Context: Dutch FMCG Distribution

The Dutch FMCG distribution sector operates under three converging pressures that make data-driven inventory management increasingly critical:

Margin compression: Major Dutch retailers — Albert Heijn, Jumbo, and Plus — have progressively tightened distributor margins. The primary lever available to mid-size distributors is operational efficiency. Excess inventory holding costs directly erode the margin available at the distributor level.

Post-pandemic supply chain volatility: Lead time unpredictability, which increased significantly during 2020–2022, has normalised at elevated levels. Distributors that relied on flat safety stock rules during this period experienced disproportionate stockout exposure. Statistical safety stock calibration — responsive to actual lead time variance — has become an operational necessity rather than a best practice.

SKU proliferation: Dutch FMCG distributors now carry more SKUs than at any point in the past decade. Managing 3,000+ SKUs across 10 locations — approximately 30,000 active SKU × Store combinations — is operationally impossible through manual monitoring. Automated risk scoring is required to direct demand planner attention to the combinations that matter.

3.2 Specific Problems at Patel Mart

Prior to DemandIQ NL, Patel Mart exhibited two simultaneous inventory failure modes:

Failure Mode	Stockouts	Overstock
Description	SKU inventory reaches zero. Retail shelf is empty. Customer purchases from competitor.	Excess inventory occupies warehouse space. Working capital is immobilised.
Primary impact	Lost revenue (permanent), emergency reorder cost, damaged retailer relationships.	Increased holding costs, product write-offs approaching expiry, reduced margins.
Affected category	Voeding (Food) — 15.64% stockout risk rate, highest of all categories.	Slow-moving SKUs across Huishouden and Hobby & Vrije Tijd.
Root cause	Safety stock insufficient for high-demand-variability SKUs under flat 14-day rule.	Safety stock excessive for low-variability SKUs under same flat 14-day rule.

3.3 Root Cause Analysis

Both failure modes share a single root cause: the flat 14-day safety stock rule. Applied uniformly across all 3,000+ SKUs, this rule does not account for:

- Demand variability — a SKU with high standard deviation in daily demand requires more buffer than a smooth-demand product.
- Lead time variance — longer or more variable lead times require larger safety stock buffers to maintain service levels.
- ABC classification — Class A SKUs (which drive 80% of revenue) warrant higher service level targets and therefore larger statistical safety stocks.
- Seasonal demand effects — Dutch public holidays (Koningsdag, Sinterklaas, Kerst) create predictable demand depressions that a flat rule cannot anticipate.

> **Financial Impact:** The 14-day blanket rule resulted in €2.37 million in excess inventory value across 23,311 SKU × Store combinations. At the standard FMCG holding cost rate of 25% (covering storage, capital, spoilage, and insurance), this translates to €593,000 in avoidable annual holding costs. This figure was computed directly from Patel Mart operational data and is reproducible via the SQL script in the project repository.

4. Methodology

4.1 Pipeline Architecture

DemandIQ NL was designed as a seven-phase pipeline, each phase producing a discrete output that feeds the next. The architecture was built on the principle that every technical decision must be traceable to a business requirement.

Phase	Name	Output	Business Purpose
1	Star Schema Design	PostgreSQL schema with staging and warehouse layers	Single source of truth for all demand and inventory queries
2	Staging Layer	58.3M rows: stg_sales, stg_prices, stg_calendar	Raw data preserved without transformation; full audit trail
3	Warehouse Layer	fact_sales (37.2M rows) + 4 dimension tables	Analysis-ready data model supporting fast BI aggregations
4	EDA & Insights	ABC classification, seasonality profile, risk heatmap	Demand structure understanding to guide model and threshold design
5	Demand Forecasting	XGBoost model; 22.7% RMSE improvement over Prophet	Accurate demand signal to underpin safety stock calculations
6	Inventory Engine	23,311-row fact_inventory with composite risk scores	Actionable stockout risk identification at SKU × Store grain
7	Power BI Dashboard	5-page executive dashboard via PostgreSQL views	Decision-ready reporting layer for demand planners and leadership

4.2 Technology Stack

The following technologies were selected based on fit with the analytical requirements, open-source availability, and compatibility with Dutch enterprise environments:

Layer	Technology	Role
Database	PostgreSQL 15	Staging, warehouse, and analytical view layer
Data Engineering	Python 3.9 (pandas, psycopg2)	ETL pipeline, chunked wide-to-long transformation
Forecasting	XGBoost 3.2.0 / Prophet 1.3.0	Demand prediction and model benchmarking

Dashboard	Power BI Desktop + Npgsql v4.0.10	Executive reporting layer; Import mode via direct PostgreSQL connection
Development	Jupyter Notebook	Interactive development, phased analysis, reproducible code

5. Findings

5.1 Data Engineering (Phases 1–3)

5.1.1 Wide-Format Transformation

The M5 dataset is distributed as a wide-format CSV file: 30,490 rows, each representing a product, with 1,913 columns representing individual sales days. PostgreSQL enforces a hard column limit of 1,600, making direct ingestion impossible.

The solution required a streaming approach: the CSV was read in configurable chunks, each chunk melted from wide to long format in memory using pandas, and the resulting long-format rows were written to PostgreSQL via psycopg2. This transformed 30,490 wide rows into 58.3 million tidy rows — one row per item, store, and date — without loading the full dataset into memory at any point.

> **Engineering Principle:** Database constraints are design signals, not obstacles. The column limit forced a data model change that produced a demonstrably superior architecture: tidy format enables fast aggregations, time-series operations, and BI tool connectivity that wide format cannot support.

5.1.2 Star Schema Design

The warehouse layer implements a standard dimensional star schema: one central fact table (fact_sales, 37.2 million rows) surrounded by four dimension tables.

Table	Grain	Key Business Content
fact_sales	Item × Store × Date	Units sold, revenue, price link
dim_date	Calendar date	Dutch public holidays, week/month/quarter flags
dim_product	SKU	Dutch category, department, NL product name
dim_store	Store location	Dutch province, city, store type
dim_price	Item × Week	Weekly sell price in euros

5.2 Exploratory Data Analysis (Phase 4)

Phase 4 produced four findings that directly shaped the design of the forecasting model and inventory engine.

5.2.1 ABC Classification

Demand-based ABC classification confirmed that 997 Class A SKUs drive 80% of total Patel Mart revenue — consistent with the Pareto principle. This finding has two operational implications: (1)

inventory intelligence resources should be concentrated on Class A SKUs first; (2) forecasting pilots should begin with Class A products to maximise impact on revenue-at-risk.

5.2.2 Dutch Seasonal Demand Patterns

Analysis of daily demand across the full five-year dataset (2020–2025) identified statistically significant demand suppression around Dutch public holidays:

- Average holiday demand suppression: −6.4% versus equivalent non-holiday periods.
- Koningsdag (27 April): consumer purchasing shifts to outdoor/leisure categories.
- Sinterklaas (5 December): significant uplift in Hobby & Vrije Tijd, followed by demand depression.
- Kerst (25–26 December): broad demand suppression across most categories.

Dutch holiday flags were embedded in dim_date and used as features in the XGBoost forecasting model. Without this, safety stock models would systematically over-order before holidays and under-order after.

5.2.3 Category Risk Profile

Voeding (Food) exhibited the highest stockout risk rate at 15.64% of SKU × Store combinations — significantly above the portfolio average of 13.2%. This is attributed to higher demand velocity, shorter acceptable stockout windows (consumers substitute food purchases immediately), and greater day-to-day demand variability relative to other categories.

5.2.4 Regional Revenue Concentration

Utrecht accounts for €21.8 million of total portfolio revenue — approximately 80% of the full €27.30 million. This concentration identified Utrecht (store CA_3) as the appropriate starting point for the forecasting pilot, maximising the business impact of model validation.

5.3 Demand Forecasting (Phase 5)

5.3.1 Model Selection Rationale

Two models were evaluated for demand forecasting on Patel Mart Class A SKU data:

Prophet (Meta, 1.3.0): An open-source additive time series forecasting tool. Prophet decomposes demand into trend, seasonality, and holiday components. It performs well on data with stable, repeating seasonal patterns and has built-in support for custom holiday calendars. However, its additive decomposition assumption limits performance on data driven by non-linear interactions between price promotions, calendar effects, and store-level demand variation.

XGBoost (3.2.0): A gradient boosted tree algorithm. XGBoost does not impose structural assumptions on the data. Given appropriately engineered features — lagged demand values, price, day-of-week, Dutch holiday flags — it learns non-linear relationships directly from the data. This makes it well-suited to FMCG demand at SKU level, where promotional effects and localised demand patterns are significant drivers.

5.3.2 Benchmark Results

Both models were evaluated on 10 Class A smooth-demand SKUs from store CA_3 Utrecht, using identical data splits, identical forecast horizons, and RMSE and MAPE as evaluation metrics.

Metric	Prophet	XGBoost	Improvement
SKUs where XGBoost wins	—	10 / 10	100%
Average RMSE	Higher	Lower	22.7% reduction
Average MAPE	Higher	Lower	21.3% reduction
Best MAPE (FOODS_3_389)	33.6%	9.9%	−23.7 percentage points
Largest improvement (MAPE)	124.3%	42.8%	FOODS_3_681

> **Finding:** XGBoost outperformed Prophet on all 10 evaluated SKUs. The performance gap was most pronounced on SKUs with complex demand interactions (e.g. FOODS_3_681, MAPE 124.3% for Prophet vs 42.8% for XGBoost), confirming that Prophet's additive seasonality assumption is not well-suited to Patel Mart FMCG demand data.

5.4 Inventory Risk Engine (Phase 6)

5.4.1 Safety Stock Methodology

Safety stock was calculated individually per SKU × Store combination using the standard statistical formula:

$$\text{Safety Stock} = Z \times \sigma \times \sqrt{L}$$

Variable	Value	Rationale
Z	1.65	95% service level. Standard for FMCG distributors; accepts 5% stockout risk.
σ	Per-SKU std deviation	Standard deviation of daily demand, calculated individually per SKU per store. Not a portfolio average.
L	Per-SKU lead time	Lead time in days, calibrated per product category. Dynamic, not hardcoded.

5.4.2 Reorder Point

The reorder point — the inventory level at which a replenishment order must be placed — was calculated as:

$$\text{Reorder Point} = (\bar{d} \times L) + \text{Safety Stock}$$

Where $\bar{d}$ is the average daily demand for the specific SKU × Store combination. This ensures that each product's reorder trigger reflects its actual consumption rate, not a portfolio-level estimate.

5.4.3 Composite Risk Score

A composite risk score (0–100) was calculated for each of the 23,311 SKU × Store combinations, combining demand variability (coefficient of variation), the ratio of current safety stock to calculated safety stock, and proximity to the reorder point. Risk tiers were calibrated against the actual score distribution:

Risk Tier	Score Range	Combinations	Recommended Action
HIGH	≥ 50	3,071 (13.2%)	Immediate replenishment review. Escalate to demand planner.
MEDIUM	25–49	—	Monitor closely. Flag for next planning cycle.
LOW	< 25	—	No immediate action required.

5.4.4 Portfolio Risk Summary

The full portfolio assessment produced the following results:

- 23,311 SKU × Store combinations assessed.
- 3,071 flagged as HIGH risk (13.2% of portfolio).
- 657 unique SKUs with some elevated risk level.
- Average safety stock across portfolio: 5.47 units per SKU × Store combination.
- Average reorder point: 17.54 units.
- Highest-risk category: Voeding (Food) at 15.64% stockout risk rate.

5.5 Business Intelligence Dashboard (Phase 7)

The Power BI dashboard was designed as a five-page executive reporting layer connected directly to PostgreSQL via five analytical views (materialized via Npgsql v4.0.10 in Import mode). Dashboard pages and their intended audiences:

Dashboard Page	Primary Audience	Key Metrics
Home	All users	Navigation, project KPI summary
Executive Summary	Senior management	€27.30M revenue, 9M units, 740 SKUs, trend chart
Stockout Risk	Demand planners	3,071 HIGH-risk combinations, 13.17% risk rate, category breakdown
Inventory Health	Operations managers	Safety stock levels, reorder points, region × category risk matrix

Regional Performance	Commercial team	Utrecht €21.8M, provincial revenue trends 2020–2025

6. Challenges & Solutions

6.1 Engineering Challenge: Wide-Format Data vs. PostgreSQL Column Limit

Challenge: The M5 sales dataset contains 30,490 rows and 1,913 day-columns — a wide-format structure that exceeds PostgreSQL's 1,600-column limit. Standard bulk ingestion approaches failed immediately.

Solution: A streaming ETL pipeline was designed that reads the CSV in configurable row chunks, melts each chunk from wide to long format in memory using pandas, and writes the resulting tidy rows to PostgreSQL incrementally via psycopg2. The pipeline included progress tracking and chunk-level error handling.

Outcome: 58.3 million tidy rows ingested without loading the full dataset into memory at any point. The resulting long-format structure is both more performant for analytical queries and more maintainable as the dataset grows.

> **Lesson:** Production-grade pipelines are defined by what happens when the naive approach fails, not by what works on clean sample data. Encountering a database constraint early forced a data model decision that produced a better architecture.

6.2 Analytical Challenge: Column Name Verification

Challenge: Across multiple phases, assumed column names (e.g., date_id, sales_qty, product_sk) did not match the actual schema, causing query failures. This was a recurring source of debugging time, particularly between Phase 5 (forecasting) and Phase 6 (inventory engine).

Solution: A consistent practice of querying information_schema.columns before writing any analytical query was established. This verification step became standard practice after the first occurrence.

Outcome: Column errors eliminated from Phase 6 onwards. The correct column names — sku_id (not product_sk), units_sold (not sales_qty), full_date (not date) — were verified before use.

6.3 Modelling Challenge: Risk Score Threshold Calibration

Challenge: The initial version of the composite risk score flagged only 6 HIGH-risk combinations from 23,311 assessed. This was clearly incorrect — a 0.03% high-risk rate does not reflect operational reality for a distributor with known stockout exposure.

Root Cause: The HIGH risk threshold (initially set at ≥ 70) was set by intuition without examining the actual score distribution. The vast majority of scores clustered below 50, making a threshold of 70 operationally meaningless.

Solution: The score distribution was analysed using percentile examination. The threshold was recalibrated from ≥ 70 to ≥ 50, and the result was validated against store volume rankings — verifying that high-revenue stores (Utrecht, Zuid-Holland) showed proportionally higher risk concentration.

Outcome: 3,071 HIGH-risk combinations identified (13.2%), consistent with operational expectations and validated against independent store-level data.

> **Lesson:** Every model output requires a sanity check against something real and independent. If a result looks implausibly clean, it almost certainly reflects a threshold or parameter problem, not genuine model accuracy.

6.4 Data Modelling Challenge: SKU Join Chain Complexity

Challenge: Joining fact_inventory to stg_prices for the business value calculation initially failed because fact_inventory.sku_id is an integer surrogate key, while stg_prices.item_id is the original M5 text identifier (e.g., FOODS_3_586). Direct casting produced null matches.

Solution: The bridge table was identified as dim_product, which holds both product_sk (integer) and item_id (text). The correct three-way join chain — fact_inventory.sku_id → dim_product.product_sk → dim_product.item_id → stg_prices.item_id — was established after verifying column names via information_schema.

Outcome: Business value calculation executed successfully, producing the €2.37M / €593K figures. The join pattern is documented and reusable for any future query requiring price enrichment.

6.5 Conceptual Challenge: Translating Technical Output into Business Value

Challenge: After Phase 6 was technically complete — risk scores calculated, safety stock validated, reorder points computed — there was no explicit financial figure summarising the project's impact. Technically correct output is not the same as a business case.

Solution: A business value quantification query (05_business_value_calculation.sql) was developed to calculate the difference between manual 14-day safety stock and statistically calibrated safety stock at SKU × Store level, multiply by unit price, and apply the 25% FMCG holding cost rate.

Outcome: €2.37M excess inventory identified. €593K annual holding cost saving quantified. These figures are reproducible, verifiable, and built entirely on Patel Mart operational data.

> **Lesson:** Business value must be calculated, not assumed. A model that produces correct outputs but cannot answer "what does this save us?" has limited organisational impact. The translation from model output to financial figure is as important as the model itself.

7. Recommendations

The following recommendations are structured across three time horizons, consistent with Dutch supply chain programme management practice: short-term operational improvements (0–3 months), medium-term capability development (3–12 months), and long-term strategic initiatives (12+ months).

7.1 Short Term (0–3 Months): Operational Implementation

7.1.1 Activate HIGH-Risk Replenishment Review Process

Immediately integrate the DemandIQ NL risk score output into the weekly demand planning cycle. The 3,071 HIGH-risk SKU × Store combinations should be reviewed by demand planners at the start of each week, prioritised by risk score and store revenue contribution. A lightweight process — daily export from the Power BI Stockout Risk page to demand planners — is sufficient to begin capturing value within days.

7.1.2 Replace the 14-Day Blanket Rule

Formally retire the flat 14-day safety stock policy. Replace it with the statistical safety stock values calculated by the DemandIQ NL inventory engine ($Z \times \sigma \times \sqrt{L}$ per SKU). This is the highest-ROI change available to Patel Mart — the engine is already built and producing results; implementation requires a procurement policy update, not additional technology work.

7.1.3 Establish Weekly Dashboard Review Cadence

Schedule a weekly 30-minute dashboard review between the data analyst, operations manager, and two demand planners. Focus on the Stockout Risk and Inventory Health pages. Document exceptions, and track how many HIGH-risk combinations are actioned each week. This builds organisational data literacy while capturing operational value.

7.2 Medium Term (3–12 Months): Capability Development

7.2.1 Extend Forecasting to Full Class A Portfolio

The Phase 5 forecasting pilot covered 10 Class A SKUs from one store. The infrastructure is in place to extend this to all 997 Class A SKUs across all 10 stores. Priority should be given to SKUs with the highest combination of revenue contribution and demand variability, as these will produce the largest safety stock accuracy improvements.

7.2.2 Write XGBoost Forecasts to PostgreSQL

The current pipeline produces XGBoost forecasts in the Jupyter notebook environment but does not write results back to PostgreSQL. Building a forecast write-back pipeline would: (1) enable the vw_forecast_vs_actual view to be populated, activating the Forecast vs Actual dashboard page; (2) allow the inventory engine to use forward-looking demand estimates rather than historical averages; (3) produce a historical forecast accuracy log for ongoing model monitoring.

7.2.3 Automate the ETL Pipeline

The current pipeline is executed manually as Jupyter notebooks. Automating the staging and warehouse refresh — using a scheduler (Apache Airflow, or a simple Python cron job on a

company server) — would enable the risk scores and dashboard to reflect current data without manual intervention. This is the step that transitions DemandIQ NL from a project into an operational system.

7.2.4 Add Supplier Lead Time Tracking

The current inventory engine uses estimated lead times per category. Replacing these with actual supplier lead time data — tracked per supplier and product — would improve the precision of both safety stock calculations and reorder point triggers. This requires integration with the procurement system and a lead time staging table, but would meaningfully improve model accuracy for SKUs with high lead time variance.

7.3 Long Term (12+ Months): Strategic Initiatives

7.3.1 Automated Replenishment Orders

Once the pipeline is automated and the model has been validated over a 6-month production period, consider building a replenishment order generation module that automatically creates purchase orders for HIGH-risk combinations whose inventory falls below the reorder point. This is the natural next step after risk scoring — moving from "flag the risk" to "resolve the risk" without human intervention for standard cases.

7.3.2 Supplier Performance Analytics

DemandIQ NL currently analyses demand. Extending the warehouse to include supplier delivery data — lead time actuals, delivery accuracy, fill rate by supplier — would enable a supplier performance module. In the Dutch FMCG context, this data is a direct input to contract negotiations and supplier development programmes.

7.3.3 ML Model Continuous Learning

As Patel Mart accumulates more sales data under DemandIQ NL operations, the XGBoost model should be retrained periodically (quarterly recommended) on the updated dataset. Implement a model versioning system so that performance can be tracked over time and model degradation can be detected before it affects operational outcomes.

7.3.4 Expand to Full Portfolio Forecasting

The current forecasting scope is Class A smooth-demand SKUs. Class B and C SKUs, and intermittent-demand products (common in Hobby & Vrije Tijd), require different modelling approaches — Croston's method or Negative Binomial regression for intermittent demand, for example. A phased expansion that addresses the full portfolio would allow safety stock calibration to be improved across all 3,000+ SKUs.

8. Conclusion

DemandIQ NL demonstrates that a mid-size Dutch FMCG distributor with existing operational data can build production-grade predictive supply chain intelligence without enterprise software procurement. The project moved from raw data ingestion to a live executive dashboard in seven phases, producing a quantified business case — €2.37 million in excess inventory, €593,000 in annual holding cost savings — built entirely from Patel Mart operational data.

Three principles governed every technical decision in this project, and are offered as guiding principles for future analytics work at Patel Mart:

- Every technical decision must be traceable to a business requirement. A star schema is not built because it is architecturally elegant — it is built because Power BI needs to render in seconds across 30,000 SKU × Store combinations.
- Model selection is a data question, not a preference. XGBoost outperformed Prophet on all 10 evaluated SKUs because FMCG demand data is non-linear and promotion-driven — not because XGBoost is universally superior.
- Business value must be calculated and documented, not assumed. The €593,000 figure is not an estimate. It is a calculation, reproducible from the project repository, that translates technical model output into a decision a senior stakeholder can act on.

The immediate priority is operational activation: replace the 14-day blanket rule with statistical safety stock, integrate the risk score into the weekly demand planning cycle, and establish a dashboard review cadence. The medium and long-term recommendations provide a roadmap for evolving DemandIQ NL from a portfolio project into a continuously operating supply chain intelligence system.

> **Business Case Summary:** Replacing the manual 14-day blanket safety stock rule with DemandIQ NL statistical safety stock modelling reduces excess inventory holding cost by an estimated €593,000 per year. The investment required for operational activation is a policy change and process update — the analytical infrastructure is already built, tested, and live.

Appendix A: Key Formulas

A.1 Safety Stock

$$SS = Z \times \sigma_d \times \sqrt{L}$$

Where Z = 1.65 (95% service level), σ_d = standard deviation of daily demand per SKU × Store, L = supplier lead time in days.

A.2 Reorder Point

$$ROP = (\bar{d} \times L) + SS$$

Where $\bar{d}$ = average daily demand per SKU × Store, L = lead time in days, SS = safety stock.

A.3 Business Value Calculation

$$\text{Excess Units} = MAX((\bar{d} \times 14) - SS, 0)$$

$$\text{Excess Value} = \text{Excess Units} \times \text{Unit Price}$$

$$\text{Annual Saving} = \text{Excess Value} \times 0.25$$

The 25% holding cost rate is the standard FMCG industry benchmark, accounting for storage, capital cost, spoilage, and insurance.

Appendix B: Project Repository

All code, SQL scripts, and documentation are available in the public GitHub repository:

GitHub: https://github.com/HP85-NL/demandiq-nl
Live Dashboard: Power BI — DemandIQ NL Dashboard

File	Description
01_star_schema.sql	Schema design and table definitions
02_staging_layer.sql	Staging table DDL
03_warehouse_layer.sql	Dimension and fact table DDL
05_business_value_calculation.sql	Business value quantification — produces €2.37M / €593K figures
07_powerbi_views.sql	Five analytical views for Power BI
04_forecasting.ipynb	Prophet vs XGBoost benchmark — all 10 Class A SKUs
05_inventory_engine.ipynb	Safety stock, reorder point, risk score computation

— End of Report —